

PATIENT MULTI-OMICS GENOMIC HEALTH SUMMARY

Patient Submitter ID: TCGA-AA-3666 | Tumor: Colon Adenocarcinoma (COAD) | Stage: Stage IIA

TCGA Project:	TCGA-COAD	cBioPortal Study:	coadread_tcg_pan_can_atlas_2018
Gender / Age:	male / 59 yrs	Vital Status / OS:	Alive (52.1 mo)
Hybrid ML Risk:	58.2% (HIGH RISK)	VQC / Classical:	54.1% / 61.0%

FOUNDATIONAL 6-DATABASE MULTI-OMICS MUTATION PROFILE (GDC, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl)

Gene	Locus (GRCh38)	Transcript	Protein Change	ClinVar Status	COSMIC ID	gnomAD Pop AF	VAF %
APC	chr5:5q22.2	ENST00000257430	p.R1450* Nonsense Truncation	Pathogenic (FAP / Sporadic CRC) VCV000000898	COSV53492104 FATHMM: 0.99 Pathogenic	0.000012	47.3%
KRAS	chr12:12p12.1	ENST00000311936	p.G12D Activating Transition	Oncogenic Driver (Anti-EGFR Resistance) VCV000012583	COSV55546255 FATHMM: 0.99 Pathogenic	0.000008	36.9%

QUANTUM VARIATIONAL ROTATION STATE ANGLES (θ_VQC / R_Y / CNOT)

Gene Symbol	Canonical Ensembl ID	VAF Hotspot Frequency	Variational Angle θ (Radians)	Qubit Hilbert Space Projection
APC	ENSG00000134982	47.3%	θ = 2.1900 rad (125.5°)	$ \psi\rangle = \cos(1.095) 0\rangle + \sin(1.095) 1\rangle$
KRAS	ENSG00000133703	36.9%	θ = 2.2150 rad (126.9°)	$ \psi\rangle = \cos(1.107) 0\rangle + \sin(1.107) 1\rangle$

PRECISION ONCOLOGY CLINICAL ACTION & PROTOCOL RECOMMENDATION

Recommended Therapeutic Pathway:
Complete mesocolic excision staging review, MSI/MMR status confirmation for immunotherapy eligibility.

Genomic Biomarker Summary: Identified somatic hotspot mutations in APC (p.R1450*), KRAS (p.G12D). Confirmed pathological classification across ClinVar and COSMIC registries with extreme somatic rarity in gnomAD exomes.

NOTICE & RESEARCH DISCLAIMER: This report is generated by the Blazeфинix Quantum-Classical Multi-Omics Research Platform using open-access academic cancer genomics datasets (NCI GDC TCGA, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl). It is calibrated strictly for computational research, translational bioinformatics, and clinical decision support benchmarking, not for standalone autonomous medical diagnosis.